Seunghyun An

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Research interests: Operating systems, chiplet-based server systems, and performance engineering.

Education

University of Wisconsin–Madison — Madison, WI, USA
Ph.D. in Computer Science — *Aug. 2024 – Present*

- GPA: 4.0/4.0
- Doctoral Minor in Astronomy

Korea Advanced Institute of Science and Technology (KAIST) — Daejeon, South Korea
B.S. in Computer Science and Electrical Engineering (Double Major) — *Mar. 2018 – Aug. 2024*

- GPA: 3.88/4.3, Magna Cum Laude
- Honor Student

Publications

cSwitch: Scheduling Linux Threads under the I/O Chiplet Wall — SOSP '26
***Seunghyun An**, Joontaek Oh, Ming Liu* code paper

- Found that Linux's conventional scheduling signals—CPU load and static topology/locality—break down on chiplet CPUs: under shared I/O-chiplet congestion, topologically local placement can run 4.3–4.5× slower than remote placement.
- Isolated this "I/O Chiplet Wall" from memory-controller saturation, showing that the slowdown originates from contention along the compute-to-I/O-chiplet path itself.
- Built cSwitch, which uses commodity performance counters to infer each thread's communication path and bandwidth demand, then schedules based on runtime path congestion rather than static topology.

Server Chiplet Networking — HotNets '25
***Seunghyun An**, Joontaek Oh, Ming Liu* paper

- Characterized communication across compute/I/O chiplets, memory, and devices on two AMD EPYC generations.
- Identified topology-dependent latency and bandwidth domains, inconsistent bandwidth-delay products, and sender-driven sharing.

Research Experience

Operating Systems for Chiplet Servers — Madison, WI
NetLab, UW-Madison (Advisor: Prof. Ming Liu) — *Aug. 2024 – Present*

- Designed controlled memory/CXL experiments and hardware-counter analysis to isolate compute–I/O chiplet contention from downstream memory-controller saturation on AMD EPYC systems.
- Implemented cSwitch's Linux `sched_ext`/eBPF path and PMU-based per-thread load attribution.
- Developed bandwidth-aware thread migration and CPU-time throttling policies.
- Investigating selective message passing for coherence-heavy inter-chiplet coordination in microkernel and multikernel designs.

Kernel-Level RPC for SmartSSD — Daejeon, South Korea
CASYS, KAIST (Advisor: Prof. Youngjin Kwon) — *Jul. 2023 – Feb. 2024*

- Ported a user-space RDMA RPC library into a Linux kernel module for SmartSSD services.
- Adapted memory management and synchronization to kernel constraints.

Concurrent Systems and File Systems in Rust — Lausanne, Switzerland
RS3Lab, EPFL (Advisor: Prof. Sanidhya Kashyap) — *Jan. 2023 – Feb. 2024*

- *MV-RLU:* Ported the protocol to Rust, preserving multi-version synchronization while confining raw-pointer operations to a small `unsafe` core.
- *ArckFS* (Trio, SOSP '23): Ported core paths from C to Rust and built an `mmap`-backed mock kernel interface.

CofferOS Physical Memory Management — Daejeon, South Korea
CASYS, KAIST (Advisor: Prof. Youngjin Kwon) — *Jul. 2022 – Dec. 2022*

- In CofferOS (later published at EuroSys '26), traced deallocation overhead to page zeroing and buddy coalescing.
- Implemented a deferred-free queue and background kernel task to move page reclamation off the deallocation path.

Honors

Presidential Science Scholarship 2018 – 2024

Dean's List, School of Engineering Spring 2021

Gold Medal and First Overall, 22nd International Astronomy Olympiad Group Beta (senior division), 2017

Technical Skills

Programming Languages: C, C++, Rust, Python

Spoken Languages: Korean (native), English (proficient)

Teaching Experience

Teaching Assistant at UW–Madison

CS 354: Machine Organization and Programming — Spring 2026/Fall 2026 (Head TA)

CS 537: Introduction to Operating Systems — Fall 2024; Spring/Fall 2025

At KAIST

Humanity/Leadership III: Hacking Class — Fall 2023

Python Tutor, Daeduk High School — 2022

Freshman Tutor: Introduction to Programming — Spring 2022

Freshman Tutor: General Physics II — Fall 2021

Professional Experience

Military Service Pyeongtaek, South Korea

KATUSA, Eighth Army Public Affairs *2019 – 2020*

- Received the U.S. Army Commendation Medal.